

1 Symmetric support of the good physical density

Theorem 1 (Actual physical symmetric support). *Let A, C be finite coordinate sets, let $\Omega \in \mathbb{C}^A$ be unit, and let $u \in (\mathbb{C}^A)^{\otimes k} \otimes \mathbb{C}^C$ be arbitrary. For an excited subset $B \subseteq \{0, \dots, k-1\}$, put $G = B^c$, $m = |G|$ and $w = (R_B \otimes I_C)u$. Enumerate the good copies by one chosen equivalence. Let $\rho_{AGC}(w)$ be the actual density obtained from $|w\rangle\langle w|$ by tracing the bad physical copies, and let P_m be the orthogonal projection onto $\text{Sym}^m \mathbb{C}^A$. Then*

$$(P_m \otimes I_C)\rho_{AGC}(w) = \rho_{AGC}(w). \quad (1)$$

The vector u need not be symmetric, normalized or nonzero. Zero copies and empty good sets are included.

This is the physical symmetric-support assertion used in [1, Section 7]. For the physical space $\mathbb{C}^Q \otimes \mathbb{C}^Y \otimes \mathbb{C}^V$, take $A = Q \times (Y \times V)$. Every middle physical coordinate is retained until the projection P_m is applied.

Proof. The actual product-density identity gives

$$\rho_{AGC}(w) = |\Omega^{\otimes m}\rangle\langle\Omega^{\otimes m}| \otimes \rho_C(w). \quad (2)$$

A permutation only reorders the identical factors of $\Omega^{\otimes m}$. Thus $P_m \Omega^{\otimes m} = \Omega^{\otimes m}$. Multiplying the product density by $P_m \otimes I_C$ proves the assertion. $\square$

References

- [1] OpenAI, *A two-dimensional area law from a global spectral gap*, September 24, 2026.